

bucex 1.2.1 — Revision Plan, Hierarchical Partial-Pooling Model, and Prior Sensitivity

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1 Package architecture overview

The `bucex` 1.2.1 package is organized into functional modules: `api/`, `components/`, `models/` (model + compiler), `observation/` (Gaussian, GEV), `priors/` (`process.py` general SD priors, `structural.py` FS hierarchical profiles), `inference/` (`config.py`, `plan.py`, `state/` with FFBS/Laplace/PGAS kernels, `fit/`), `core/` (`FitResult`), `diagnostics/`, `datasets/` (Uccle), `io/` (checksummed `.bucex` archives), `plotting/`, and `simulate/`.

The inference interface has three orthogonal axes:

$$\text{parameterization} \in \{\text{centered}, \text{disturbance}, \text{fruehwirth_schnatter}\}, \quad (1)$$

$$\text{engine} \in \{\text{ffbs}, \text{laplace}, \text{pgas}\}, \quad (2)$$

$$\text{prior profile} \in \{\text{manuscript_lasso}, \text{regularized_lasso}, \text{regularized_horseshoe}, \text{pc}, \text{norm}\} \quad (3)$$

A single `FitResult` object stores chain-preserving outputs with array conventions

$$\text{state_draws} \in \mathbb{R}^{C \times D \times (T+1) \times d_\alpha}, \quad (4)$$

$$\text{parameter_draws}, \text{plan}, \text{sampler_diagnostics}, \text{auxiliary_draws} \text{ as companion objects.} \quad (5)$$

The Uccle dataset contains six monthly series from 1892–2022 (1572 observations each): `TXm` and `TNm` (Gaussian monthly means of daily max/min), `TXx` and `TNx` (GEV upper-tail monthly maxima), and `TXn` and `TNn` (GEV lower-tail monthly minima, sign-flipped internally). The v1.2.1 release validation reported 39 passing tests, finite state arrays of shape $(1, 1, 1573, 13)$ for all six full-length fits, and for PGAS/FS `TXx` a particle count of 24, path-change rate 1.0, and median minimum particle ESS 12.61.

2 Revision plan: pseudo-Laplace to PGAS

The revision direction is to replace or supplement the iterated-Laplace pseudo-observation GEV engine with PGAS (Particle Gibbs with Ancestor Sampling) as the primary reported method. Iterated-Laplace is approximate because it linearizes the GEV score/Hessian around a current state estimate and embeds this Gaussian pseudo-observation inside FFBS. PGAS instead

uses a conditional SMC with ancestor sampling kernel that is exact-invariant for the target posterior [11, 12].

Before making PGAS the headline result, the observed median minimum particle ESS of 12.61/24 should be stress-tested by scaling particle count and reporting ESS degradation over the full time axis, not only a summary median. The revision should include direct Laplace-versus-PGAS comparisons on identical series/prior settings, alongside wall-clock trade-offs. A practical split is to keep Laplace for fast approximate prior-sensitivity screening while reserving PGAS for confirmatory final inference.

3 The base model as a state-space model (per series)

For series i and month $t = 1, \dots, T$, define latent predictor

$$\eta_{i,t} = Z'_t \alpha_{i,t}. \quad (6)$$

The observation model is

$$y_{i,t} \mid \eta_{i,t}, \theta_i \sim \begin{cases} \mathcal{N}(\eta_{i,t}, \sigma_i^2), & \text{for TXm, TNm,} \\ \text{GEV}(\eta_{i,t}, \sigma_i, \xi_i), & \text{for TXx, TXn, TNx, TNn.} \end{cases} \quad (7)$$

The latent state recursion is linear-Gaussian:

$$\alpha_{i,t+1} = T_t \alpha_{i,t} + R_t \zeta_{i,t}, \quad (8)$$

$$\zeta_{i,t} \sim \mathcal{N}(0, Q_i), \quad (9)$$

with block-diagonal T_t and R_t assembled from local-linear-trend, dummy-seasonal, and optional regression components as in `bucex/models/compiler.py`. For GEV observations this is a dynamic generalized linear model [5, 6], specialized to extreme-value dynamics [8, 9, 7, 10]. The FS parameterization is an algebraic reparameterization of the same state recursion using signed scale coefficients [4].

4 Adding partial pooling: hierarchical layer across the six series

Partial pooling introduces a hyperprior layer without altering within-series state dynamics (hierarchical DLM structure [27, 28]). For GEV shape pa-

rameters on four series ($i = 1, \dots, 4$):

$$\xi_i \mid \mu_\xi, \tau_\xi \sim \mathcal{N}(\mu_\xi, \tau_\xi^2), \quad (10)$$

$$\mu_\xi \sim \mathcal{N}(0, s_0^2), \quad (11)$$

$$\tau_\xi \sim \text{Half-Cauchy}(0, A) \quad (\text{or Half-Normal}(0, A)). \quad (12)$$

Trend-rate pooling by series type $g(i) \in \{\text{mean}, \text{extreme}\}$ can be written

$$\beta_i \mid \mu_\beta^{(g(i))}, \tau_\beta \sim \mathcal{N}(\mu_\beta^{(g(i))}, \tau_\beta^2), \quad (13)$$

where the key climate estimand is

$$\Delta_\beta = \mu_\beta^{(\text{extreme})} - \mu_\beta^{(\text{mean})}, \quad (14)$$

reported with a credible interval.

A full posterior factorization is

$$p(\mu_\xi, \tau_\xi, \{\alpha_i, \xi_i, \sigma_i, Q_i\}_{i=1}^6 \mid \{y_i\}_{i=1}^6) \propto p(\mu_\xi) p(\tau_\xi) \prod_{i=1}^6 \left[p(\xi_i \mid \mu_\xi, \tau_\xi) p(\sigma_i) p(Q_i) p(\alpha_{i,0}) \right. \quad (15)$$

$$\left. \times \prod_{t=1}^T p(\alpha_{i,t+1} \mid \alpha_{i,t}, Q_i) p(y_{i,t} \mid \alpha_{i,t}, \sigma_i, \xi_i) \right]. \quad (16)$$

Implementation can remain post-hoc hierarchical: conditional on (μ_ξ, τ_ξ) , each series likelihood factorizes exactly as now, so FFBS/PGAS stay per-series and unchanged. Add one Gibbs step per sweep:

$$\mu_\xi \mid \{\xi_i\}, \tau_\xi \sim \mathcal{N}\left(\frac{\tau_\xi^{-2} \sum_i \xi_i}{n\tau_\xi^{-2} + s_0^{-2}}, \left(n\tau_\xi^{-2} + s_0^{-2}\right)^{-1}\right), \quad (17)$$

plus Metropolis or slice sampling for τ_ξ . This requires no changes to `compiler.py/CompiledModel`; pooling lives in the prior graph. Hierarchical GEV and regional-frequency precedents include [23, 24, 25, 26, 21].

5 Prior choice discussion, especially for ξ

FS structural SSVS uses discrete component-inclusion spike-and-slab indicators (trend/seasonal active, static, dynamic), whereas general process-SD spike-and-slab continuously shrinks innovation SDs toward zero [18, 19].

Regularized horseshoe [15, 16] and SSVS are complementary: horseshoe yields smooth shrinkage factors κ_j , while SSVS yields interpretable posterior inclusion probabilities. Reporting both is recommended.

For the GEV shape parameter ξ , current choices (Uniform($-0.5, 0.5$) or truncated Normal($0, 1$)) can over-weight heavy-tail behavior near $|\xi| \approx 0.5$ relative to expected temperature-extreme regimes (often $|\xi| < 0.2$). A better default is

$$\xi \sim \mathcal{N}(0, \sigma_\xi^2) \mathbf{1}_{(-0.5, 0.5)}(\xi), \quad \sigma_\xi \in [0.2, 0.3], \quad (18)$$

or a PC prior shrinking toward the Gumbel base model ($\xi = 0$) with explicit tail-probability calibration [20]. Prior sensitivity should compare Uniform, truncated Normal, and PC- ξ via decision-relevant outputs (50- and 100-year return levels), and extend `prior_sensitivity.py` to include `ssvs`. Summaries should emphasize `period_rate_summary` and `static_summary` rather than raw parameter tables.

6 Mean/max dependence problem

For each month t , TXm and TXx are deterministic functionals of shared daily values $X_{t,d}$ (and similarly TNm, TNx, and TNn are all deterministic functionals of the same month- t daily minima path):

$$\text{TXm}_t = \frac{1}{n_t} \sum_{d=1}^{n_t} X_{t,d}, \quad (19)$$

$$\text{TXx}_t = \max_{d=1, \dots, n_t} X_{t,d}. \quad (20)$$

Independent per-series fitting imposes zero contemporaneous residual/innovation covariance, which is generally false and can understate joint uncertainty and invalidate naive tests of trend differences.

Candidate fixes, increasing in invasiveness:

1. **SUTSE / correlated innovations across series.** Use correlated disturbances and, for mixed Gaussian/GEV pairs, a Gaussian-copula residual link:

$$u_{\text{TXm},t} = \Phi\left(\frac{y_{\text{TXm},t} - \eta_{\text{TXm},t}}{\sigma_{\text{TXm}}}\right), \quad (21)$$

$$u_{\text{TXx},t} = F_{\text{GEV}}(y_{\text{TXx},t}; \eta_{\text{TXx},t}, \sigma_{\text{TXx}}, \xi_{\text{TXx}}), \quad (22)$$

$$\left(\Phi^{-1}(u_{\text{TXm},t}), \Phi^{-1}(u_{\text{TXx},t})\right) \sim \mathcal{N}_2(\mathbf{0}, R_t), \quad R_t = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}. \quad (23)$$

2. **Shared dynamic factor model.** Introduce

$$\eta_{i,t} = \mu_{i,t} + \lambda_i f_t, \quad (24)$$

$$f_{t+1} = T_f f_t + w_t, \quad w_t \sim \mathcal{N}(0, Q_f), \quad (25)$$

to test common climate forcing in mean and extreme series.

3. **Full daily-process model.** Model daily temperatures directly and derive monthly mean/max as induced functionals.

A concrete revision recommendation is to implement (1) first (low overhead, one extra ρ per mean/max pair), then layer (2) if time allows.

7 Cross-validation approach

Standard random k -fold CV is invalid because it breaks temporal dependence. Use leave-future-out / expanding-window CV: fit on 1892– T , forecast ahead, score, then roll T forward [36]. Tail-aware scoring should prioritize CRPS or quantile/pinball loss at high quantiles (90th/95th/99th) over RMSE [37]. Engine and prior comparisons (PGAS vs Laplace, alternative prior profiles) should use this same blocked protocol, aligned across all six series, to support fair hierarchical/joint model comparisons.

8 Main paper contribution points (summary)

1. Unified inference framework in **bucex**: one model grammar with exactness-labeled engines and orthogonal parameterization/engine/prior choices.
2. PGAS as exact-invariant benchmark to validate or replace historical pseudo-Laplace approximations, especially near tail events.
3. Systematic prior sensitivity (especially ξ) linked to return-level consequences.
4. Leave-future-out CV with tail-appropriate scores to justify inference settings empirically.
5. Partial pooling across six Uccle series (hierarchical ξ and trend-rate contrasts), including explicit treatment of mean/max dependence, to support valid climate-trend claims.

References

- [1] Harvey, A.C. (1989). *Forecasting, Structural Time Series Models and the Kalman Filter*.
- [2] West, M. & Harrison, J. (1997). *Bayesian Forecasting and Dynamic Models* (2nd ed.).
- [3] Durbin, J. & Koopman, S.J. (2012). *Time Series Analysis by State Space Methods* (2nd ed.).
- [4] Frühwirth-Schnatter, S. & Wagner, H. (2010). Stochastic model specification search for Gaussian and partial non-Gaussian state space models. *Journal of Econometrics*.
- [5] West, M., Harrison, J. & Migon, H.S. (1985). Dynamic generalized linear models and Bayesian forecasting. *JASA*.
- [6] Fahrmeir, L. & Tutz, G. (1994). *Multivariate Statistical Modelling Based on Generalized Linear Models*.
- [7] Huerta, G. & Sansó, B. (2007). Time-varying models for extreme values. *Environmental and Ecological Statistics*.
- [8] Nogaj, M., Yiou, P., Parey, S., Malek, F. & Naveau, P. (2006). Amplitude and frequency of temperature extremes over the North Atlantic region. *GRL*.
- [9] Parey, S., Malek, F., Laurent, C. & Dacunha-Castelle, D. (2007). Trends and climate evolution: statistical approach for very high temperatures in France. *Climatic Change*.
- [10] Rootzén, H. & Katz, R.W. (2013). Design Life Level: quantifying risk in a changing climate. *Water Resources Research*.
- [11] Andrieu, C., Doucet, A. & Holenstein, R. (2010). Particle Markov chain Monte Carlo methods. *JRSS-B*.
- [12] Lindsten, F., Jordan, M.I. & Schön, T.B. (2014). Particle Gibbs with Ancestor Sampling. *JMLR*.
- [13] Chopin, N. & Papaspiliopoulos, O. (2020). *An Introduction to Sequential Monte Carlo*.

- [14] Yu, Y. & Meng, X.-L. (2011). To center or not to center: interweaving techniques for Bayesian data augmentation (ASIS). *JCGS*.
- [15] Carvalho, C.M., Polson, N.G. & Scott, J.G. (2010). The horseshoe estimator for sparse signals. *Biometrika*.
- [16] Piironen, J. & Vehtari, A. (2017). Sparsity information and regularization in the horseshoe and other shrinkage priors. *Electronic Journal of Statistics*.
- [17] Park, T. & Casella, G. (2008). The Bayesian lasso. *JASA*.
- [18] George, E.I. & McCulloch, R.E. (1993). Variable selection via Gibbs sampling. *JASA*.
- [19] George, E.I. & McCulloch, R.E. (1997). Approaches for Bayesian variable selection. *Statistica Sinica*.
- [20] Simpson, D., Rue, H., Riebler, A., Martins, T.G. & Sørbye, S.H. (2017). Penalizing model component complexity: a principled, practical approach to constructing priors. *Statistical Science*.
- [21] Martins, E.S. & Stedinger, J.R. (2000). Generalized maximum-likelihood generalized extreme-value quantile estimators for hydrologic data. *Water Resources Research*.
- [22] Coles, S. (2001). *An Introduction to Statistical Modeling of Extreme Values*.
- [23] Coles, S. & Tawn, J.A. (1996). A Bayesian analysis of extreme rainfall data. *Applied Statistics*.
- [24] Cooley, D., Nychka, D. & Naveau, P. (2007). Bayesian spatial modeling of extreme precipitation return levels. *JASA*.
- [25] Renard, B. (2011). A Bayesian hierarchical approach to regional frequency analysis. *Water Resources Research*.
- [26] Sang, H. & Gelfand, A.E. (2009). Hierarchical modeling for extreme values observed over space and time. *Environmental and Ecological Statistics*.
- [27] Gamerman, D. & Migon, H.S. (1993). Dynamic hierarchical models. *JRSS-B*.

- [28] Gelman, A. & Hill, J. (2007). *Data Analysis Using Regression and Multilevel/Hierarchical Models*.
- [29] Renard, B. & Lang, M. (2007). Use of a Gaussian copula for multivariate extreme value analysis: some case studies in hydrology. *Advances in Water Resources*.
- [30] Genest, C. & Favre, A.-C. (2007). Everything you always wanted to know about copula modeling but were afraid to ask. *Journal of Hydrologic Engineering*.
- [31] Sang, H. & Genton, M.G. (2014). Tapered composite likelihood for spatial max-stable models. *Spatial Statistics*.
- [32] Katz, R.W., Parlange, M.B. & Naveau, P. (2002). Statistics of extremes in hydrology. *Advances in Water Resources*.
- [33] Fawcett, L. & Walshaw, D. (2006). A hierarchical model for extreme wind speeds. *JRSS-C*.
- [34] Aguilar, O. & West, M. (2000). Bayesian dynamic factor models and portfolio allocation. *Journal of Business & Economic Statistics*.
- [35] Stock, J.H. & Watson, M.W. (2016). Dynamic factor models, factor-augmented VARs, and SVARs in macroeconomics. In *Handbook of Macroeconomics*.
- [36] Bürkner, P.-C., Gabry, J. & Vehtari, A. (2020). Approximate leave-future-out cross-validation for Bayesian time series models. *Journal of Statistical Computation and Simulation*.
- [37] Gneiting, T. & Raftery, A.E. (2007). Strictly proper scoring rules, prediction, and estimation. *JASA*.
- [38] Harvey, A.C. & Koopman, S.J. (1997). Multivariate structural time series models. In *System Dynamics in Economic and Financial Models*.